

Pragmatic Intent Detection for AI Assistants

An Empirical Study of Conversational Context and Linguistic Ambiguity

Machine & Deep Learning Project Proposal

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1. Project Concept & Main Idea

Modern AI assistants are often evaluated on whether they can understand the literal content of a user's request, yet everyday conversation frequently depends on pragmatic meaning: what a speaker intends beyond the words that are explicitly stated. A user may say, "It's getting cold in here," when the intended request is to increase the temperature, or "I have an interview tomorrow and I have barely practiced," when the underlying intent is to ask for preparation help. These examples are difficult because the same utterance can function differently depending on prior dialogue, the task being discussed, and how indirectly the user expresses the request.

This project will investigate how conversational context and degree of linguistic ambiguity affect a model's ability to detect and interpret pragmatic user intent. The primary research question is: How do conversational context and linguistic ambiguity interact when machine learning and language models infer a user's underlying intent? The project will treat this primarily as an empirical comparative study rather than as an attempt to invent a completely new neural architecture. The goal is to measure where current approaches succeed, where they fail, and how much additional context helps as requests become less explicit.

The system will use a two-stage formulation. Stage 1 will perform indirect-intent detection: given a user utterance and, when available, prior dialogue, the model will determine whether the user is expressing an indirect or implied request. Stage 2 will perform pragmatic interpretation: for utterances classified as indirect, the system will infer a concise direct form of the user's likely underlying request. The study will use a controlled mixed benchmark covering approximately six to eight common assistant-intent categories, such as information seeking, recommendation, planning/scheduling, troubleshooting, navigation, advice/support, clarification, and action/device requests.

Two experimental variables will be controlled. First, examples will vary in indirectness, from direct requests to conventionally indirect phrasing, strongly implied requests, and requests whose intended meaning is only clear from context. Second, models will receive different amounts of dialogue context, such as the current utterance alone, one prior turn, or a larger multi-turn history. This makes it possible to test whether context-aware models degrade less sharply than utterance-only models as pragmatic inference becomes more demanding. The project will compare a classical NLP baseline (for example, TF-IDF with logistic regression), a transformer classifier such as RoBERTa, an instruction-tuned LLM approach, and a two-stage hybrid pipeline that combines classification with generated pragmatic interpretation.

The expected contribution is primarily empirical: a quantitative account of how context and indirectness influence pragmatic-intent performance across model families and intent categories. A secondary contribution will be a small, controlled benchmark built from existing dialogue resources plus carefully designed or LLM-assisted examples that are manually reviewed. Rather than claiming that synthetic data alone represents natural conversation, the dataset design will emphasize contrastive cases, provenance, human validation, and reproducibility. The final analysis will include

classification metrics, generation-quality evaluation, ablations over context length, and error analysis of ambiguous or incorrectly inferred intents.

2. Relevant Resources & Background

Takayama, Kajiwar, and Arase (2021), DIRECT: Direct and Indirect Responses in

Conversational Text Corpus [1]. DIRECT provides more than 71,000 indirect-direct utterance pairs with multi-turn dialogue history derived from MultiWOZ. It is especially relevant because it operationalizes a closely related task: mapping indirect conversational language to an explicit intended meaning.

George and Mamidi (2020), Conversational Implicatures in English Dialogue: Annotated

Dataset [2]. This work provides context, utterance, and manually annotated implicated meanings for conversational examples. It demonstrates a useful annotation structure for separating literal utterances from inferred pragmatic meaning.

Mannekote et al. (2024), Making Task-Oriented Dialogue Datasets More Natural by

Synthetically Generating Indirect User Requests [3]. This work introduces IndirectRequests and an LLM-assisted pipeline for producing realistic indirect user requests. It is directly relevant to the proposed hybrid dataset strategy and to the problem of evaluating smaller assistant models on indirect language.

Żelasko, Pappagari, and Dehak (2021), What Helps Transformers Recognize Conversational

Structure? [4]. Their analysis shows that broader conversational context can help transformer models disambiguate dialogue acts, motivating the proposed context ablations and the hypothesis that pragmatic intent cannot always be inferred from an isolated utterance.

Arai and Ren (2026), DRInQ: Evaluating Conversational Implicature with Controlled Context

Variation [5]. DRInQ evaluates pragmatic reasoning while systematically varying context, making it a recent and highly relevant example of controlled experimental design for conversational implicature.

Research gap. Prior work has examined indirect responses, implicatures, dialogue acts, and indirect task-oriented requests, but the proposed study focuses on the interaction between two controlled factors—amount of conversational context and degree of indirectness—across several assistant-intent categories and multiple model families. The project will also evaluate both detection (is the intent indirect?) and interpretation (what is the intended request?), rather than treating pragmatic understanding as a single classification problem. This creates a focused undergraduate research contribution: not a claim to solve general pragmatics, but a reproducible empirical study of when context becomes necessary for intent inference and which modeling approaches are most robust as requests become more implicit.

3. 10-Week Execution Plan

	Milestone / Deliverable
k 1	Finalize research questions, intent taxonomy, indirectness levels, dataset schema, repository structure, and evaluation plan.
k 2	Collect and inspect existing dialogue resources (e.g., DIRECT, conversational implicature data, IndirectRequests). Define licensing/provenance notes and select usable subsets.
k 3	Create the custom controlled benchmark. Generate candidate direct/indirect contr pairs, manually review them, balance intent categories, and define train/dev/test sp
k 4	Perform exploratory data analysis and quality checks. Create annotation guideline

	validate a subset for ambiguity/label consistency and revise problematic cases.
k 5	Implement baseline models: TF-IDF + logistic regression and an utterance-only transformer classifier. Establish Stage 1 indirect-intent metrics.
k 6	Implement context-aware transformer experiments with zero, one, and multiple pr turns. Compare performance across indirectness levels and intent categories.
k 7	Implement LLM prompting and the two-stage hybrid pipeline. Evaluate Stage 2 pragmatic interpretations using structured labels plus semantic/human review on a sample.
k 8	Run main experiments, context ablations, and model comparisons. Record precision, recall, F1, confusion matrices, and interpretation-quality results.
k 9	Conduct statistical/error analysis. Inspect failure cases, ambiguous examples, context sensitivity, and whether particular intent categories are systematically harder.
k 10	Finalize figures/tables, write the technical report, clean the GitHub repository for reproducibility, document limitations/ethics, and prepare the presentation/demo.

Selected References

- [1] J. Takayama, T. Kajiwar, and Y. Arase, “DIRECT: Direct and Indirect Responses in Conversational Text Corpus,” Findings of EMNLP, 2021.
- [2] E. J. George and R. Mamidi, “Conversational Implicatures in English Dialogue: Annotated Dataset,” Procedia Computer Science, vol. 171, pp. 2316–2323, 2020.
- [3] A. Mannekote, J. Nam, Z. Li, J. Gao, K. E. Boyer, and B. J. Dorr, “Making Task-Oriented Dialogue Datasets More Natural by Synthetically Generating Indirect User Requests,” arXiv:2406.07794, 2024.
- [4] P. Želasko, R. Pappagari, and N. Dehak, “What Helps Transformers Recognize Conversational Structure? Importance of Context, Punctuation, and Labels in Dialog Act Recognition,” Transactions of the ACL, vol. 9, pp. 1163–1179, 2021.
- [5] H. J. Arai and X. Ren, “DRInQ: Evaluating Conversational Implicature with Controlled Context Variation,” Proceedings of ACL, 2026.